TITLE PAGE

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Good day, respected panel members, faculty, and fellow students.

I am Dether John Gorre, a Master of Science in Information Technology student. Today, I will present my proposed study entitled "Development of an AI-Driven Academic Burnout Detection and Early Warning System Using a Multi-Factor Burnout Index for College Students."

This study focuses on utilizing Artificial Intelligence and Machine Learning to predict and monitor burnout risks among college students through continuous assessment and early warning mechanisms.

BACKGROUND OF THE STUDY

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Academic burnout has become one of the major challenges experienced by college students today.

Students are exposed to various academic pressures such as examinations, projects, deadlines, and heavy workloads. Over time, these pressures may result in emotional exhaustion, mental fatigue, reduced motivation, and poor academic performance.

Because of these effects, burnout has become an important issue that educational institutions need to address.

BACKGROUND OF THE STUDY

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Although many studies have investigated burnout, most existing approaches focus on individual factors such as stress or workload separately.

Current monitoring methods are often survey-based and do not provide continuous monitoring or early warning capabilities.

As a result, there is a need for a more comprehensive and intelligent approach that can monitor multiple burnout-related factors and provide timely intervention.

LITERATURE MAP

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This literature map summarizes the major areas that support the proposed study.

The reviewed literature covers academic burnout, stress, workload, sleep behavior, study routines, Artificial Intelligence, Machine Learning, and early warning systems.

These studies helped identify the variables, technologies, and research gaps that guided the development of the proposed framework.

SYNTHESIS AND GAPS

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Based on the reviewed studies, burnout is influenced by multiple factors including stress, workload, sleep behavior, and study routines.

Researchers have also demonstrated that Artificial Intelligence and Machine Learning can improve prediction and monitoring systems.

However, most studies analyze these variables separately rather than collectively, limiting their ability to provide a comprehensive burnout assessment.

SYNTHESIS AND GAPS

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The review also revealed several limitations in existing studies. Most systems rely on survey-based assessment and focus on single-variable analysis.

Furthermore, current approaches lack integrated burnout prediction models and AI-driven early warning capabilities.

These limitations highlight the need for a more comprehensive burnout monitoring framework.

SYNTHESIS AND GAPS

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To address these gaps, this study proposes the development of an AI-Driven Academic Burnout Detection and Early Warning System.

The proposed system integrates stress level, academic workload, sleep hours, and study time through a Multi-Factor Burnout Index or MFBI.

Machine Learning algorithms are then applied to predict burnout risk and support early intervention.

COMPARISON OF FEATURES

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This table compares the proposed study with existing studies.

While previous studies provide stress monitoring and limited AI integration, they generally lack sleep monitoring, study time monitoring, multi-factor frameworks, and early warning functionality.

The proposed system combines all these features into a unified burnout prediction framework, which represents the primary contribution of the study.

NOVELTY OF THE STUDY

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The novelty of this study lies in the development of the Multi-Factor Burnout Index or MFBI.

Unlike existing studies that focus on individual variables, the MFBI integrates stress level, academic workload, sleep hours, and study time into a single burnout assessment framework.

The study also combines Machine Learning prediction, continuous monitoring, and AI-driven early warning notifications to improve burnout detection among students.

STATEMENT OF THE PROBLEM

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This study aims to address the challenge of monitoring and predicting academic burnout among college students.

Specifically, the study seeks to assess burnout-related variables, develop a Multi-Factor Burnout Index, apply Machine Learning algorithms for classification, develop an AI-driven early warning system, and evaluate the performance of the developed models.

These questions serve as the foundation for the proposed system.

OBJECTIVES OF THE STUDY

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The general objective of the study is to develop an AI-Driven Academic Burnout Detection and Early Warning System using the Multi-Factor Burnout Index.

To achieve this objective, the study will assess burnout-related variables, develop the MFBI framework, apply Random Forest and Decision Tree algorithms, develop the system, and evaluate its performance using standard Machine Learning metrics.

THEORETICAL FRAMEWORK

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The study is anchored on three theories: Conservation of Resources Theory, Stress Theory, and Social Cognitive Theory.

These theories explain how academic demands, stress exposure, and behavioral factors contribute to burnout development.

Together, they provide theoretical support for the inclusion of stress level, workload, sleep hours, and study time in the proposed framework.

THEORETICAL FRAMEWORK

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Collectively, these theories explain why burnout should be examined using multiple factors rather than a single variable

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They support the idea that burnout develops through the interaction of academic pressure, resource depletion, stress exposure, and behavioral patterns.

This theoretical foundation strengthens the rationale for developing the Multi-Factor Burnout Index.

CONCEPTUAL FRAMEWORK

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This slide presents the conceptual framework of the study.
The inputs consist of stress level, academic workload, sleep hours, and study time.

These variables undergo preprocessing, MFBI computation, and Machine Learning analysis to generate burnout classification and early warning notifications.

SCOPE AND LIMITATION

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The study focuses on selected college students in Lanao del Norte.

It utilizes stress level, workload, sleep hours, and study time as predictors of burnout and applies Random Forest and Decision Tree algorithms for analysis.

The system is intended solely for burnout monitoring, prediction, and early warning and is not designed for clinical diagnosis.

SIGNIFICANCE OF THE STUDY

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The study benefits students by promoting burnout awareness and early intervention.

It also assists teachers and guidance counselors in monitoring students who may be vulnerable to burnout.

Furthermore, the study contributes to educational monitoring and serves as a reference for future AI-based burnout research.

RESEARCH METHODOLOGY

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This study uses a Developmental Research Design to develop and evaluate the proposed system.

A Quantitative Research Approach will be used to collect and analyze burnout-related data.

The study also follows Agile Methodology to ensure continuous development and improvement of the system.

RESEARCH INSTRUMENTS

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The study will utilize validated instruments to collect burnout-related data.

These include the Perceived Stress Scale, workload monitoring forms, sleep logs, and study time monitoring sheets.

These instruments will provide the data needed for burnout assessment and prediction.

DATA GATHERING PROCEDURE

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The data gathering process begins with participant registration and consent.

Students will undergo weekly monitoring for stress, workload, sleep hours, and study time.

The collected data will then be prepared for Machine Learning analysis and burnout prediction.

DATA NORMALIZATION

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Before computing the burnout index, all variables will be normalized to a common scale.

Normalization ensures that all variables contribute fairly during analysis.

For sleep hours, reverse normalization is applied because lower sleep duration corresponds to higher burnout risk.

MULTI-FACTOR BURNOUT INDEX (MFBI)

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The Multi-Factor Burnout Index, or MFBI, is the core contribution of this study.

It combines stress level, academic workload, sleep hours, and study time into a single burnout score.

This score will serve as the basis for burnout prediction and classification.

SYSTEM DEVELOPMENT METHODOLOGY

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The proposed system will be developed using Agile Methodology.

The process includes planning, analysis, design, development, testing, deployment, and maintenance.

This approach allows continuous improvement throughout system development.

MACHINE LEARNING ALGORITHMS

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The study will utilize Random Forest and Decision Tree algorithms.

These algorithms will analyze burnout-related variables and classify students according to their burnout risk level.

The goal is to identify students who may require intervention at an early stage.

EVALUATION METRICS

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To evaluate model performance, the study will use Accuracy, Precision, Recall, and F1-Score.

These metrics will determine how effectively the Machine Learning models predict and classify burnout risk.

DATA FLOW DIAGRAM LEVEL 0

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This diagram presents the overall flow of information within the proposed system.

It shows how students, teachers, and guidance counselors interact with the burnout detection and early warning system.

DATA FLOW DIAGRAM LEVEL 1

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This diagram provides a more detailed view of the system processes.

It illustrates how data is collected, processed, analyzed, and transformed into burnout predictions and early warning notifications.

ENTITY RELATIONSHIP DIAGRAM

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This Entity Relationship Diagram presents the database structure of the proposed system.

It shows the relationships among students, monitoring records, burnout results, and system users.

SYSTEM FLOWCHART

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This flowchart illustrates the overall operation of the system.

The process starts with data collection, followed by MFBI computation, Machine Learning prediction, burnout classification, and early warning notification.

ETHICAL CONSIDERATIONS

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The study will ensure voluntary participation, informed consent, and confidentiality of all collected information.

The system is intended only for burnout monitoring and prediction and not for clinical diagnosis.